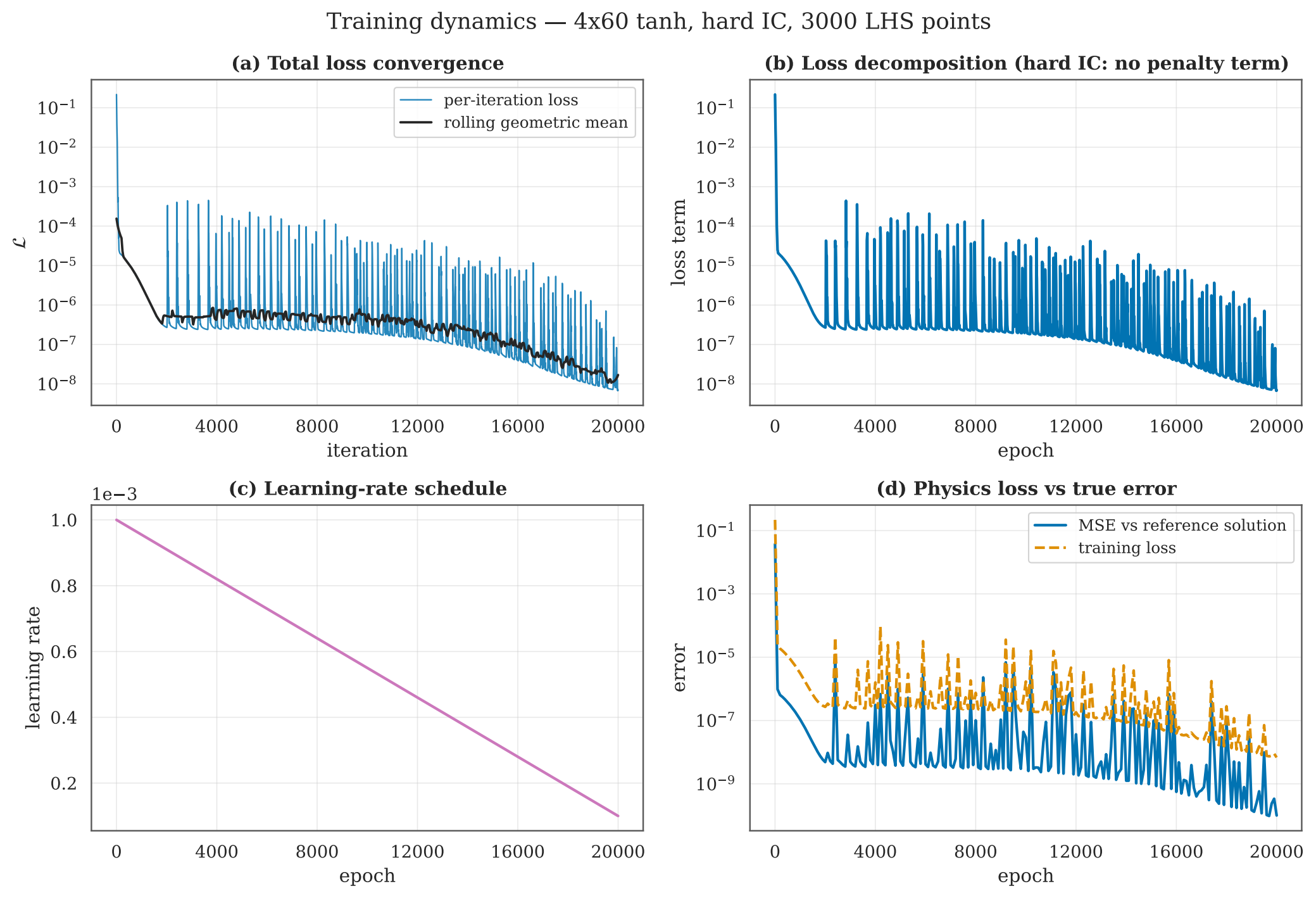

Training dynamics — 4x60 tanh, hard IC, 3000 LHS points
(a) Total loss convergence
per-iteration loss
rolling geometric mean
$\mathcal{L}$
iteration
(b) Loss decomposition (hard IC: no penalty term)
loss term
epoch
(c) Learning-rate schedule
1e−3
learning rate
epoch
(d) Physics loss vs true error
MSE vs reference solution
training loss
error
epoch